



1. There is a positive integer n such that $n\%$ of n is 4. What is $(n^2)\%$ of n ?

- (A) 10 (B) 16 (C) 20 (D) 80 (E) 1600

2. A group of k children are playing with a 52-card deck. The deck is split so that every child receives a different number of cards, and each child receives at least one card. What is the maximum possible value of k ?

- (A) 8 (B) 9 (C) 10 (D) 11 (E) 12

3. What is the sum of the digits of the greatest positive integer n such that $8n + 9$ is divisible by $2n - 5$?

- (A) 3 (B) 8 (C) 9 (D) 10 (E) 15

4. Josh wants to buy trading cards for 30 cents. He has a bag of pennies with radius 1, nickels with radius 3, dimes with radius 5, and quarters with radius 8. Using these coins, $A\pi$ is the maximum total area of the coins that can be used to purchase the trading cards. What is the sum of the digits of A ?

- (A) 7 (B) 8 (C) 10 (D) 12 (E) 15

5. A positive integer is called a palindrome if it reads the same forwards and backwards. For example, 121 and 777 are palindromes. How many three-digit palindromes are divisible by 6?

- (A) 10 (B) 12 (C) 13 (D) 14 (E) 18

6. In regular hexagon $ABCDEF$, let $P_1, P_2, P_3, P_4, P_5, P_6$ be the midpoints of $\overline{AB}, \overline{BC}, \overline{CD}, \overline{DE}, \overline{EF}, \overline{FA}$, respectively. Lines P_1P_2, P_3P_4, P_5P_6 are the sides of a triangle δ . What is the ratio of the area of δ to the area of $\triangle ACE$?

- (A) $\frac{3}{2}$ (B) $\frac{27}{16}$ (C) 2 (D) $\frac{9}{4}$ (E) 3

7. Suppose $(x \heartsuit y) = xy - \frac{x}{y}$ for all nonzero y . What is the product of all values of n satisfying $(n \heartsuit 3) = n \cdot (3 \heartsuit n)$?

- (A) -3 (B) $-\frac{8}{3}$ (C) -1 (D) $\frac{8}{9}$ (E) 1

8. What is the sum of the squares of the real parts of the roots of $x^2 - 2\sqrt{5}x + (5 - 8i)$?

- (A) 1 (B) $2\sqrt{5}$ (C) 10 (D) 18 (E) 19

9. Let $\triangle ABC$ have $AB = 13$, $BC = 14$ and $AC = 15$. Moreover, let O be the circumcenter of $\triangle ABC$ and let D be the foot of the altitude from A to BC . What is the area of $\triangle AOD$?

- (A) 12 (B) 14 (C) 16 (D) 18 (E) 20

10. Isosceles right triangle $\triangle ABC$ with right angle at B has area 1. Let M be the midpoint of $\overline{AB}$. A line ℓ passing through M and perpendicular to AC intersects segment AC at X and the extension of line BC at Y . What is the area of triangle $\triangle CXY$?

- (A) 1 (B) $\frac{3\sqrt{2}}{4}$ (C) $\frac{3+\sqrt{2}}{4}$ (D) $\frac{9}{8}$ (E) $\frac{11+6\sqrt{2}}{16}$

11. Let $a_1, a_2, \dots, a_n$ be a sequence of positive integers such that no two terms are relatively prime. Suppose the product of all the terms of the sequence is $2^1 \cdot 3^2 \cdot 5^3 \cdot 7^4 \cdot 11^5$. What is the maximum possible value of n ?

- (A) 6 (B) 11 (C) 15 (D) 16 (E) 21

12. Define the *content* of a nonnegative rational number as its denominator when written in simplest form. In other words, n is the content of $\frac{m}{n}$, where m and n are relatively prime positive integers. Suppose p and q are rational numbers such that the content of p is 5040 and the content of q is 3960. How many possible values are there for the content of $p + q$?

(A) 1 (B) 6 (C) 10 (D) 18 (E) 24

13. Oliver selects two nonnegative integers a and b less than 1000 uniformly and at random, and attempts to compute their sum. However, he completely forgets to carry over; so, for instance, to add 13 and 19, he writes down the units digit, 2, discards the carryover, and writes the sum of the tens digits, 2 (thus his answer is 22). Suppose that the positive difference between his answer and the correct answer is D . What is the expected value of D ?

(A) 450 (B) 495 (C) 499 (D) $\frac{999}{2}$ (E) 500

14. Let ABC be a triangle with area 1, and let M , N , P be the midpoints of $\overline{BC}$, $\overline{CA}$, $\overline{AB}$, respectively. A point X is chosen uniformly and at random in the interior of $\triangle ABC$. What is the expected value of the area of the convex hull of $\{M, N, P, X\}$?

(A) $\frac{5}{16}$ (B) $\frac{1}{3}$ (C) $\frac{11}{32}$ (D) $\frac{3}{8}$ (E) $\frac{1}{2}$

15. Let $r_1 < r_2 < r_3$ be the roots of the cubic $x^3 - 3x - 1$ and let $s_1 < s_2 < s_3$ be the roots of the cubic $x^3 - 3x + 1$. What is the value of $|(r_1 + s_1)(r_2 + s_2)(r_3 + s_3)|$?

(A) 0 (B) 1 (C) 3 (D) 6 (E) 9

16. Trapezoid $ABCD$ with $\overline{AB} \parallel \overline{CD}$ is inscribed in a circle ω . The tangent to ω at A intersects line CD at P , and the tangent to ω at C intersects line AB at Q . Lines AP and CQ intersect at T . Suppose that $DA = AB = BC = 5$ and $CD = 10$. What is $TP \cdot TQ$?

(A) 50 (B) 60 (C) 75 (D) 100 (E) 150

17. For a n -minute call, where n is a positive real number, a phone line charges $n^2 - 3n + 4.41$ dollars. If the minimum cost of a series of phone calls that total to 17 minutes is expressed as $\frac{a}{b}$, where a and b are relatively prime positive integers, what is the sum of digits of a ?

(A) 3 (B) 13 (C) 15 (D) 16 (E) 19

18. A deck of 8 distinct cards are laid in a stack. In each *shuffling*, a card can be chosen from the stack, removed, and placed at the top of the stack. How many stacks of the cards are there for which a minimum of 4 shufflings are needed to produce this stack from the original stack?

(A) 224 (B) 336 (C) 630 (D) 1344 (E) 1680

19. Let $ABCD$ be a convex quadrilateral such that $AB = 4$, $BC = 4$, $CD = 3$, $DA = 7$. There exists a unique point P inside quadrilateral $ABCD$ such that the areas of $\triangle PAB$, $\triangle PBC$, $\triangle PCD$, $\triangle PDA$ are all numerically equal. What is the value of $PA^2 + PB^2 + PC^2 + PD^2$?

(A) 30 (B) 35 (C) 40 (D) 45 (E) 50

20. How many ordered pairs of real numbers (a, b) satisfy the equations $\lceil a^4 \rceil - 2\lfloor a^2 \rfloor b + \lceil b^2 \rceil^2 = 0$, $\lceil a^2 \rceil + 2a\lfloor b^2 \rfloor + \lceil b \rceil^2 = 0$?

(A) 1 (B) 2 (C) 3 (D) 4 (E) 8

21. Let $\triangle ABC$ be an isosceles triangle with side lengths $AB = AC = 8$, and let ω and I be the incircle and incenter of $\triangle ABC$, respectively. Suppose ω is tangent to BC , AC , AB at D , E , F , respectively. Segment AI intersects ω at X . Given that lines XE , FI , BC intersect at a common point, then the length of BC is $\frac{m}{n}$, where m and n are relatively prime positive integers. What is the value of $m + n$?

(A) 17 (B) 18 (C) 19 (D) 20 (E) 21

22. What is the remainder when $(1^{2020} + 2^{2020} + \dots + 2019^{2020})^{2020}$ is divided by 2020?

(A) 0 (B) 1 (C) 404 (D) 1616 (E) 2019

23. In acute $\triangle ABC$, let D and E be the feet of the altitudes from A to $\overline{CB}$ and C to $\overline{AB}$, respectively. Suppose that $AB = 4$, $CB = 3$, and $DE = 2$. The sum of all possible values of the length of $\overline{AC}$ can be written in the form $\frac{\sqrt{p+q}}{r}$, where p is not divisible by the square of any prime. What is the value of $p + q + r$?

(A) 72 (B) 74 (C) 75 (D) 78 (E) 92

24. Let N be the number of integer sequences $a_1, a_2, \dots, a_{10}$ such that for every pair (i, j) , where $1 \leq i \leq j \leq 10$, $|a_i + a_{i+1} + \dots + a_j| \leq 2$. What is the remainder when N is divided by 100?

(A) 0 (B) 25 (C) 47 (D) 49 (E) 99

25. How many monic polynomials $P(x)$ of degree 12, whose coefficients are integers between 0 and 12, inclusive, have the property that either $P(x) + 1$ or $P(x) - 1$ is divisible by 13 for every integer x ?

(A) 792 (B) 924 (C) 1001 (D) 1716 (E) 3003

Time limit: 75 minutes